# India's Agricultural Crop Production Analysis

| TEAM ID | SWTID-2026-4021 |
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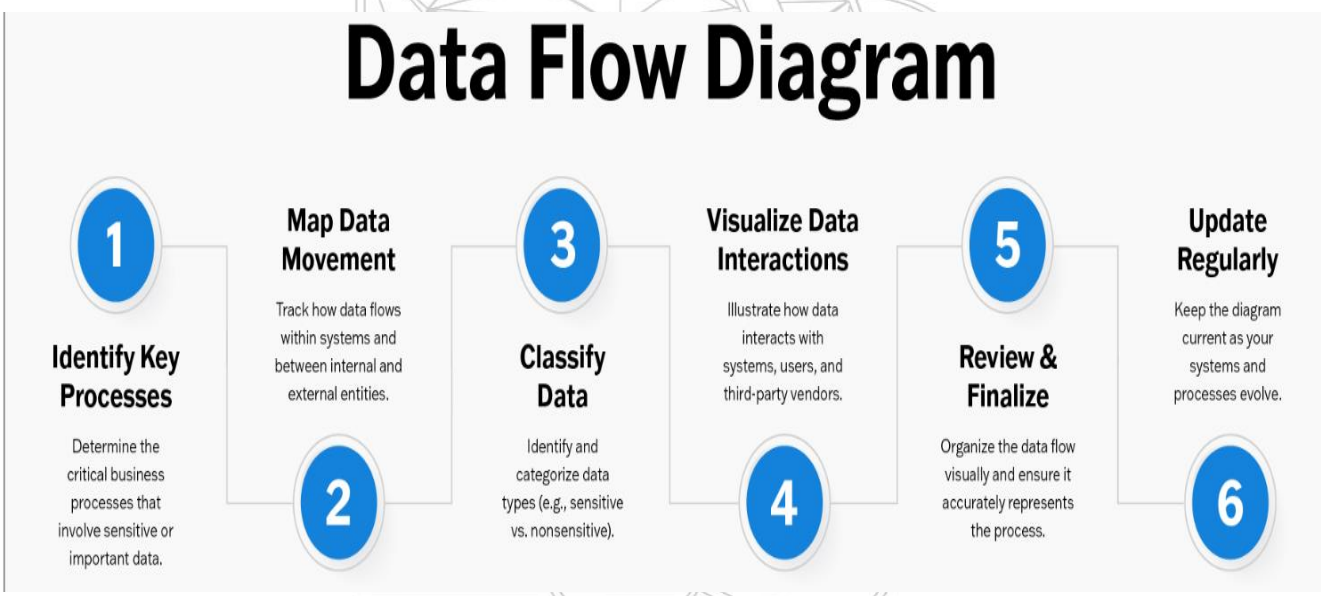
| PROJECT TITLE | India's Agricultural Crop Production Analysis |
| MAX MARKS | 3 MARKS |

## 3.3 – Data Flow Diagram




India's Agricultural Crop Production Analysis – Data Flow Diagram (DFD)

Process Description

### 1. Data Collection Module

Collects agricultural data such as crop type, cultivated area, rainfall, soil conditions, irrigation details, and market prices.

### 2. Data Analysis

Processes and analyzes historical and current crop production data to identify trends and productivity patterns.

### 3. Weather & Market Analysis

Evaluates weather conditions, rainfall forecasts, and market price fluctuations that influence crop production.

### 4. Yield Prediction & Forecasting

Uses analytical models to estimate future crop yields, production levels, and potential agricultural risks.

### 5. Report & Dashboard Generation

Generates reports, charts, and recommendations for farmers, agricultural experts, and government agencies.

## Inputs

- Crop Production Data
- Soil Information
- Weather Data
- Irrigation Records
- Market Price Data

## Outputs

- Crop Yield Predictions
- Production Trend Reports
- Weather Impact Analysis
- Market Analysis Reports
- Farmer Recommendations
- Policy Decision Support Reports

## Conclusion

The Data Flow Diagram illustrates how agricultural data is collected, processed, analyzed, and transformed into meaningful insights. The system supports farmers and policymakers by providing accurate crop production analysis, yield forecasting, and decision-making assistance for sustainable agricultural development in India.